Personalised Lifestyle Advice: A Semantic Modelling Approach

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1. Introduction

This research aims to develop a model that integrates heterogeneous lifestyle and biomedical data sources to support the structured generation of personalised lifestyle advice. It focuses on the connection between personal biomedical data with scientifically proven lifestyle advice. The use cases focus on type 2 diabetes, prediabetes and cardiovascular disease. The symptoms of these diseases can be substantially reduced through lifestyle interventions. [1] Currently, the knowledge required to compose appropriate lifestyle changes is stored as lists of triggers and corresponding advice. In practice, patient data are manually compared to these triggers to determine which advice may be applicable. The process of generating lifestyle advice is currently time-consuming. Semantic modelling allows lifestyle advice to be generated in a structured and interoperable form. [2] It enables linking between measurements, behaviours, protocols, and interpretations.

2. Conceptual Model

The conceptual model was developed through an iterative design process that combined insights from scientific literature on lifestyle adaptations with the practical expertise of a lifestyle coach. This collaborative approach ensured that the model captured both the biomedical data related to the disease and the behavioural components involved in lifestyle change. The complete conceptual model is illustrated in Figure 1. In this model, the Person serves as the central entity from which all relevant lifestyle and biomedical information flows. Each person is associated with one or more Measurements, which may include biomedical indicators (such as blood pressure, cholesterol, weight, height, or HbA1c levels) or (self-reported) lifestyle-related observations. Every measurement is collected through a Measurement Event, capturing contextual information such as the date and time the measurement was taken. Each measurement event follows a Measurement Protocol, ensuring data is gathered consistently and in an interpretable way. Measurements are linked to a corresponding Measurement Interpretation Protocol, which defines the rules or thresholds (such as upper and lower bounds) used to interpret raw values. These interpretation protocols enable the derivation of Indications, which may indicate diagnostic cues, risk levels, or other relevant interpretations. Lifestyle behaviour is represented through a structured hierarchy. Reported Lifestyle behaviour describes the individual's current behavioural patterns, while Target Lifestyle behaviour specifies the behaviours the individual aims to achieve as part of a lifestyle intervention. Both fall under the broader category of Lifestyle behaviour, which is divided into five domains: Exercising LSB, Stress Management LSB, Substance Use LSB, Diet LSB, and Sleep LSB. Each domain contains multiple attributes; for example, sleep behaviour includes aspects such as sleep quality, average sleep duration, nap patterns, and chronotype. Using this information, a Lifestyle Change Plan can be created. This plan outlines the sequence of behavioural changes the individual should follow, based on their current behavioural profile and the indications derived from

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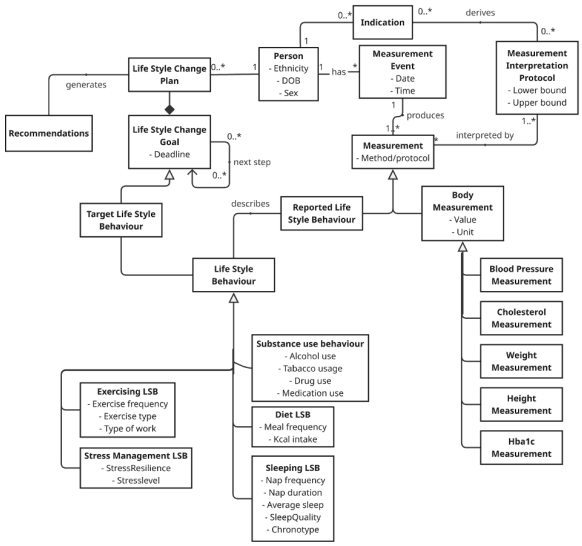

Figure 1: Conceptual model mapping an individual's biomedical and lifestyle data

their biomedical measurements. Each lifestyle change plan consists of one or more Lifestyle Change Goals, representing actionable steps like increasing physical activity, improving sleep regularity, or reducing substance use, and may include deadlines or other constraints. These goals connect the person's behavioural data, biomedical measurements, and personalised recommendations.

3. Preliminary Work

To evaluate if the model is representational, a mock dataset was created. This dataset contains realistic personal data, the type of advice a lifestyle coach would offer is also included in this dataset. The mock data populates all the relevant properties in the model. Using Python and rdflib [3], these mock data points are turned into mock individuals.

4. Future Work

A separate python script will retrieve data from the knowledge base and generate lifestyle advice based on the information encoded in the model. The advice produced by this script will be compared with the advice provided by a lifestyle coach. This comparison will serve as a validation step to assess whether the model captures all relevant data points required for generating appropriate lifestyle advice.

5. Conclusion

This work presents an semantic model for integrating biomedical measurements, lifestyle behaviour, and interpretation protocols in support of personalised lifestyle advice. The conceptual model demonstrates how biomedical and lifestyle data can be structured in a interoperable way. Future validation will make clear how accurate the advice will be.

6. Declaration on the Use of Generative AI

Generative Artificial Intelligence tools were used to assist with wording and language refinement of the proposal. During the research phase, AI tools were used in a supportive role (e.g., for clarification or exploration). All research activities, analysis, interpretation of sources, and conclusions were conducted by the authors. The authors take full responsibility for the content of this contribution.

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